

UNBOUNDED NEGATIVE CURVATURE OPERATOR IN A COLLAPSING CIRCLE QUOTIENT OF S^4

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. Consider the unit round sphere

$$S^4 \subset \mathbb{C}^2 \oplus \mathbb{R}$$

with the effective isometric circle action of weights 1 and 2. Form the quotient metric on $S^4 \times_{S^1} (\epsilon S^1)$. At either fixed pole, we compute the complete curvature operator on $\Lambda^2 T_p S^4$. With $t = \epsilon^{-2}$, its smallest eigenvalue is

$$1 - 2t = 1 - \frac{2}{\epsilon^2}.$$

It follows that the curvature operator has no lower bound independent of ϵ as the circle factor collapses. The sectional curvatures at the fixed point remain nonnegative; the negative direction is represented by a nondecomposable two-form, illustrating the distinction between sectional curvature and the curvature operator in dimension four.

1. THE QUOTIENT METRIC

Write

$$S^4 = \{(z_1, z_2, r) \in \mathbb{C}^2 \oplus \mathbb{R} : |z_1|^2 + |z_2|^2 + r^2 = 1\}$$

with the unit round metric g . Let S^1 act by

$$\lambda \cdot (z_1, z_2, r) = (\lambda z_1, \lambda^2 z_2, r).$$

This action is isometric and effective. Its fixed set consists of the two poles

$$p_{\pm} = (0, 0, \pm 1).$$

Give the auxiliary circle $S^1 = \mathbb{R}/(2\pi\mathbb{Z})$ the metric $\epsilon^2 d\theta^2$, and let S^1 act diagonally on $S^4 \times \epsilon S^1$. Translation on the second factor makes the diagonal action free. The quotient is diffeomorphic to S^4 by

$$[x, z] \mapsto z^{-1} \cdot x.$$

Let K be the infinitesimal action field on S^4 . For a tangent vector Y , the horizontal lift is obtained by minimizing

$$|Y + sK|_g^2 + \epsilon^2 s^2$$

over $s \in \mathbb{R}$. The minimum occurs at

$$s = -\frac{\langle Y, K \rangle}{|K|^2 + \epsilon^2}.$$

Consequently the quotient metric is

$$g_{\epsilon}(Y, Z) = g(Y, Z) - \frac{\langle Y, K \rangle \langle Z, K \rangle}{|K|^2 + \epsilon^2}.$$

Putting

$$t = \epsilon^{-2}$$

gives the equivalent expression

$$g_t = g - \frac{t}{1 + t|K|^2} K^{\flat} \otimes K^{\flat}. \quad (1)$$

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2. THE FIXED-POINT TWO-JET

Work at $p = p_+$. Choose the positively oriented round-orthonormal basis

$$e_1 = \partial_{x_1}, \quad e_2 = \partial_{y_1}, \quad e_3 = \partial_{x_2}, \quad e_4 = \partial_{y_2},$$

where $z_j = x_j + iy_j$. The infinitesimal isotropy

$$A = (\nabla K)_p$$

satisfies

$$Ae_1 = e_2, \quad Ae_2 = -e_1, \quad Ae_3 = 2e_4, \quad Ae_4 = -2e_3.$$

Use g -normal coordinates obtained from $\exp_p$. Equivariance of the exponential map gives $K(x) = Ax$ in these coordinates. Since

$$g_{ij} = \delta_{ij} + O(|x|^2), \quad K(0) = 0, \quad |K|^2 = O(|x|^2),$$

the quadratic part of the metric perturbation $q = g_t - g$ is

$$q_{ij} = -t(Ax)_i(Ax)_j + O(|x|^4). \quad (2)$$

The denominator correction in (1) starts in degree four and does not affect the curvature at p .

Adopt the convention

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z, \quad R_{ijkl} = \langle R(e_i, e_j)e_l, e_k \rangle.$$

The unit round sphere then has curvature operator I_{Λ^2} . Since $q(0) = dq(0) = 0$, the curvature increment at p is

$$\Delta R_{ijkl} = \frac{1}{2} (q_{jk,il} - q_{jl,ik} - q_{ik,jl} + q_{il,jk}). \quad (3)$$

Lemma 2.1. *The nonzero independent entries of the curvature increment are*

$$\begin{aligned} \Delta R_{1212} &= 3t, & \Delta R_{3434} &= 12t, \\ \Delta R_{1234} &= 4t, & \Delta R_{1324} &= 2t, & \Delta R_{1423} &= -2t. \end{aligned}$$

Proof. Differentiate (2) twice:

$$q_{ij,k\ell} = -t(A_{ik}A_{j\ell} + A_{i\ell}A_{jk}).$$

Substitution into (3) gives the displayed values. They satisfy the first Bianchi identity

$$\Delta R_{1234} - \Delta R_{1324} + \Delta R_{1423} = 0.$$

□

3. BIANCHI POLARIZATION

Let

$$\alpha = e^{12} + 2e^{34}.$$

For a decomposable unit two-form $\eta = X \wedge Y$, the fixed-point sectional curvature is

$$\sec_{g_t}(X, Y) = 1 + 3t\langle \alpha, \eta \rangle^2.$$

The rank-one operator $3t\alpha \otimes \alpha$ has the same quadratic values on decomposable two-forms but fails the first Bianchi identity. With the selected orientation,

$$*e^{12} = e^{34}, \quad *e^{13} = -e^{24}, \quad *e^{14} = e^{23}.$$

The required four-form correction is $-2t*$.

Proposition 3.1. *At either fixed pole, the curvature operator is*

$$\mathcal{R}_t = I + 3t\alpha \otimes \alpha - 2t*.$$

Proof. The right-hand side has the entries in Lemma 2.1. The Hodge-star term vanishes on decomposable quadratic evaluations because $\langle *\eta, \eta \rangle = 0$, while its alternating contribution restores the Bianchi identity. □

4. THE COMPLETE OPERATOR AND ITS SPECTRUM

In the ordered orthonormal basis

$$(e^{12}, e^{13}, e^{14}, e^{23}, e^{24}, e^{34})$$

of $\Lambda^2 T_p S^4$, Proposition 3.1 gives

$$\mathcal{R}_t = \begin{pmatrix} 1+3t & 0 & 0 & 0 & 0 & 4t \\ 0 & 1 & 0 & 0 & 2t & 0 \\ 0 & 0 & 1 & -2t & 0 & 0 \\ 0 & 0 & -2t & 1 & 0 & 0 \\ 0 & 2t & 0 & 0 & 1 & 0 \\ 4t & 0 & 0 & 0 & 0 & 1+12t \end{pmatrix}. \quad (4)$$

Theorem 4.1. *The smallest eigenvalue of the curvature operator at either fixed pole is*

$$1 - 2t = 1 - \frac{2}{\epsilon^2}.$$

In particular, the curvature operators of $S^4 \times_{S^1} (\epsilon S^1)$ are not bounded below uniformly as $\epsilon \rightarrow 0$.

Proof. The eigenvalues of the (e^{12}, e^{34}) block are

$$1 + \frac{15 + \sqrt{145}}{2}t, \quad 1 + \frac{15 - \sqrt{145}}{2}t.$$

The remaining two 2-by-2 blocks have eigenvalues $1 + 2t$ and $1 - 2t$, each with multiplicity two. Since $15 - \sqrt{145} > 0$, the smallest eigenvalue is $1 - 2t$.

Equivalently, the unit two-form

$$\omega_- = \frac{e^{13} - e^{24}}{\sqrt{2}}$$

is an exact eigenvector:

$$\mathcal{R}_t \omega_- = (1 - 2t) \omega_-.$$

As $t = \epsilon^{-2}$, the eigenvalue tends to $-\infty$. □

Remark 4.2. There is no contradiction with the nonnegative sectional-curvature values at the fixed point. The two-form ω_- is not decomposable. In dimension four, a lower bound for the curvature operator is strictly stronger than a lower bound for sectional curvature.

5. SCOPE

The example is compact, the circle action is effective and isometric, and the diagonal quotient is smooth for every $\epsilon > 0$. It therefore answers negatively the unrestricted question of whether the curvature operator remains uniformly bounded below in this collapsing construction. The argument concerns the curvature operator, not merely one sectional curvature, and the divergence occurs at a fixed point of codimension four.

APPENDIX A. CHECKLIST FOR VERIFICATION

- (1) Derive the quotient metric (1) by horizontal minimization and verify $t = \epsilon^{-2}$.
- (2) Check the infinitesimal isotropy matrix with weights 1 and 2.
- (3) Verify that only the quadratic term in (2) contributes at the fixed point.
- (4) Recompute the curvature entries from (3) and check the first Bianchi identity.
- (5) Verify the polarized formula $I + 3t \alpha \otimes \alpha - 2t*$.
- (6) Diagonalize the three 2-by-2 blocks in (4).
- (7) Check directly that $(e^{13} - e^{24})/\sqrt{2}$ has eigenvalue $1 - 2t$.

REFERENCES

- [1] American Institute of Mathematics, *Flows in Riemannian geometry*, workshop problem list and summary, September 16–20, 2019.